SOC Home Lab

Part 3 – Deployment of Wazuh Virtual Machine and agent on Windows Endpoint

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Technologies Used

- **Oracle VirtualBox**
- **Windows 11 Microsoft Sysmon**
- **Wazuh**

OBJECTIVE:

The objective of this lab is to install and deploy a Wazuh virtual machine so the endpoint telemetry, on the Windows 11 Victim virtual machine, can be collected in a centralized location for monitor and analysis.

Lab Setup

Wazuh virtual machine image was downloaded, installed, and deployed on Oracle VirtualBox. During Wazuh start-up, administrator credentials were inputted and verified, while logged in as administrator Wazuh dashboard was retrieved. Powered on Windows 11 virtual machine, deployed Wazuh on Windows endpoint via command.

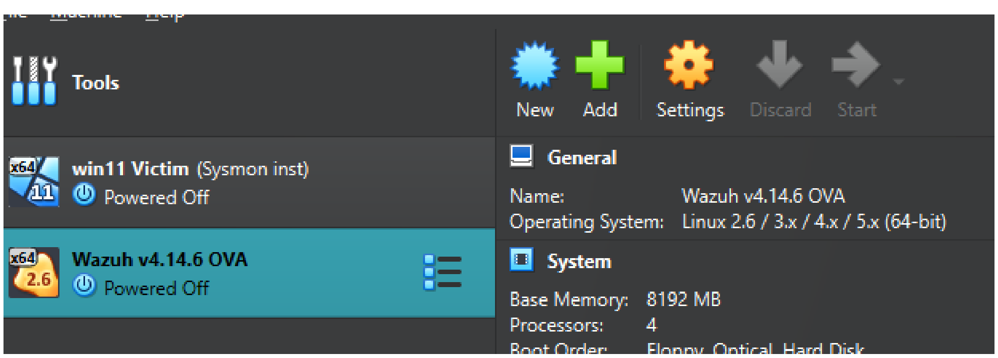
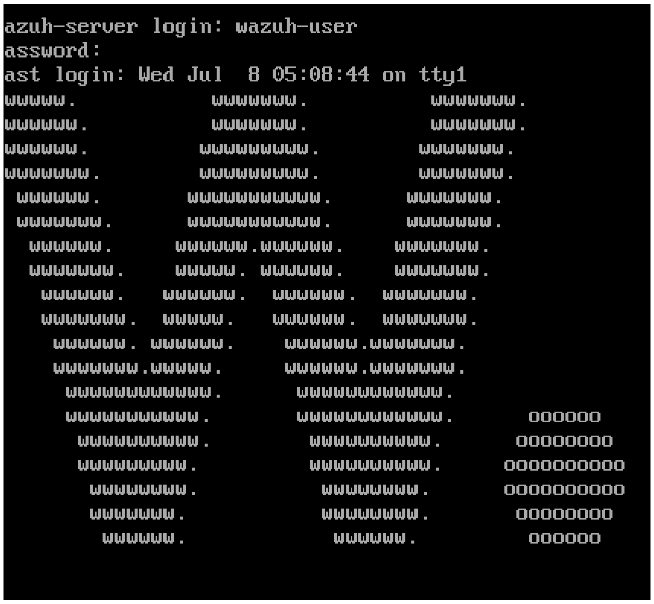


Figure 1. Successful installation of Wazuh on VirtualBox



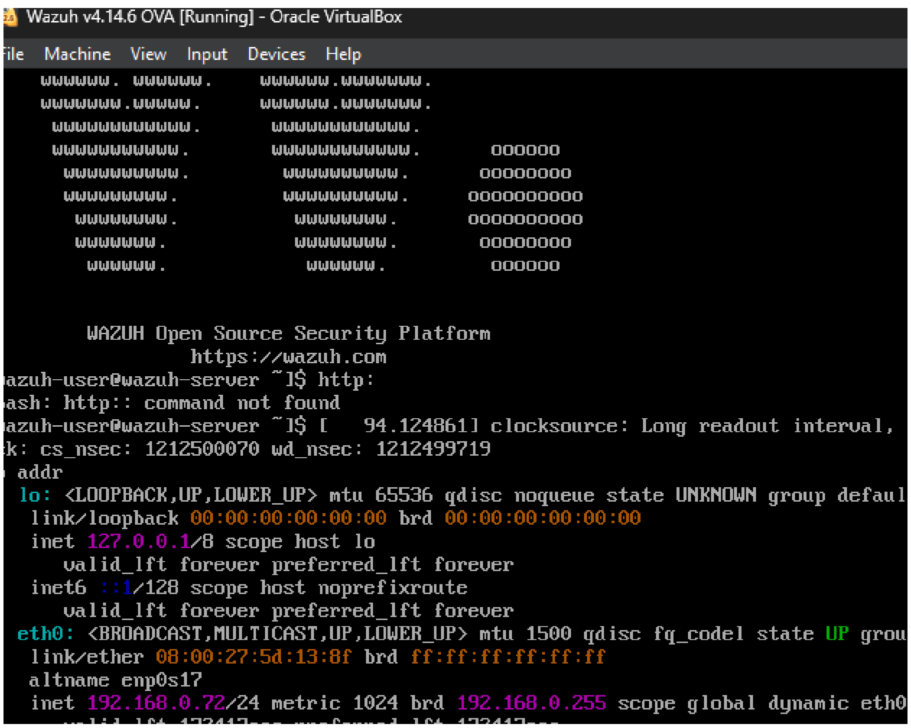


Figure 1. Wazuh virtual appliance successfully deployed in Oracle VirtualBox with network configuration verified.

Figure 2. Wazuh dashboard authentication login

Troubleshooting

Problem

The Wazuh Dashboard displayed:

> API connections could be down or inaccessible.

The dashboard reported the Wazuh API status as Offline.

Investigation

Verified:

- Network connectivity
- VirtualBox adapter configuration
- Wazuh Manager service status

Initially suspected a networking issue because the Windows endpoint could not deploy the Wazuh agent.

Resolution

Restarted the Wazuh Manager

“sudo systemctl restart Wazuh- manager”

Verified:

“sudo systemctl status Wazuh-manager”

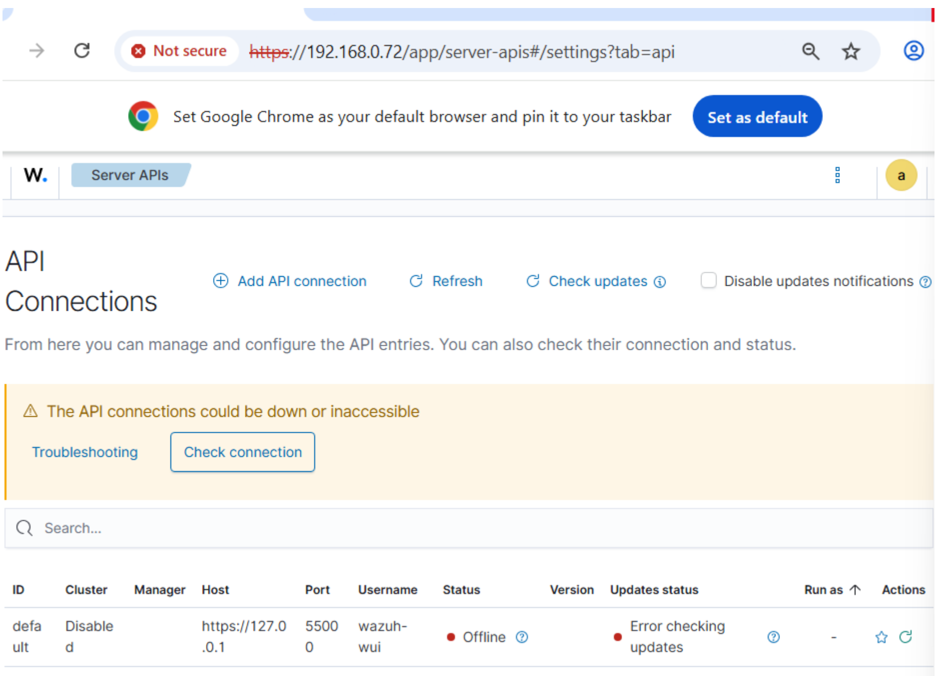


Figure 3. Wazuh Dashboard displays API connection Status as down and inaccessible, indicating that Dashboard is unable to talk to Wazuh API.

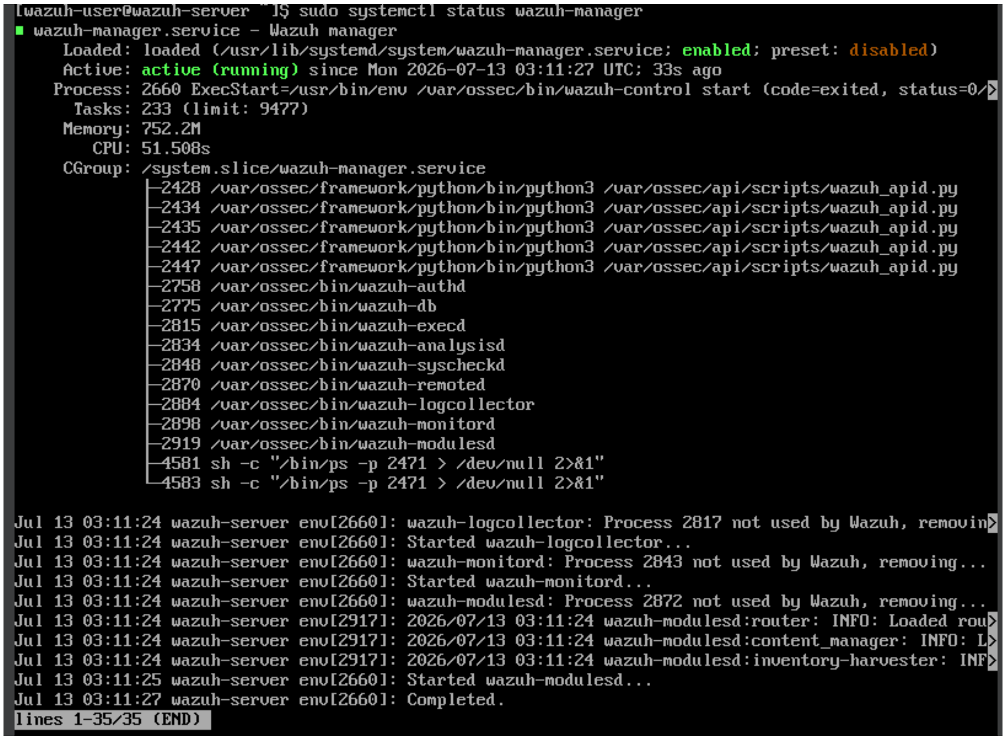


Figure 4. Displays Wazuh managers online and enable after restart, initially Wazuh server displayed manager API was offline.

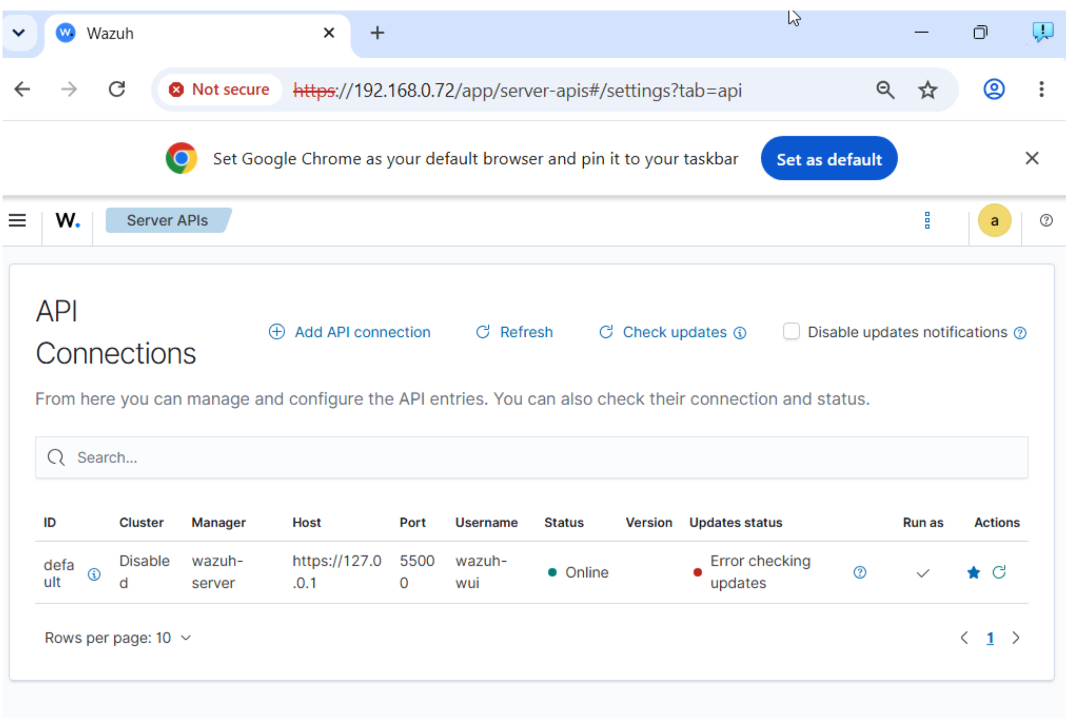


Figure 5. Successfully displays Wazuh manager and API back to online and available status after troubleshooting and restarting on Wazuh server. Now will be able to access API and deploy Wazuh agent on Windows 11 VM.

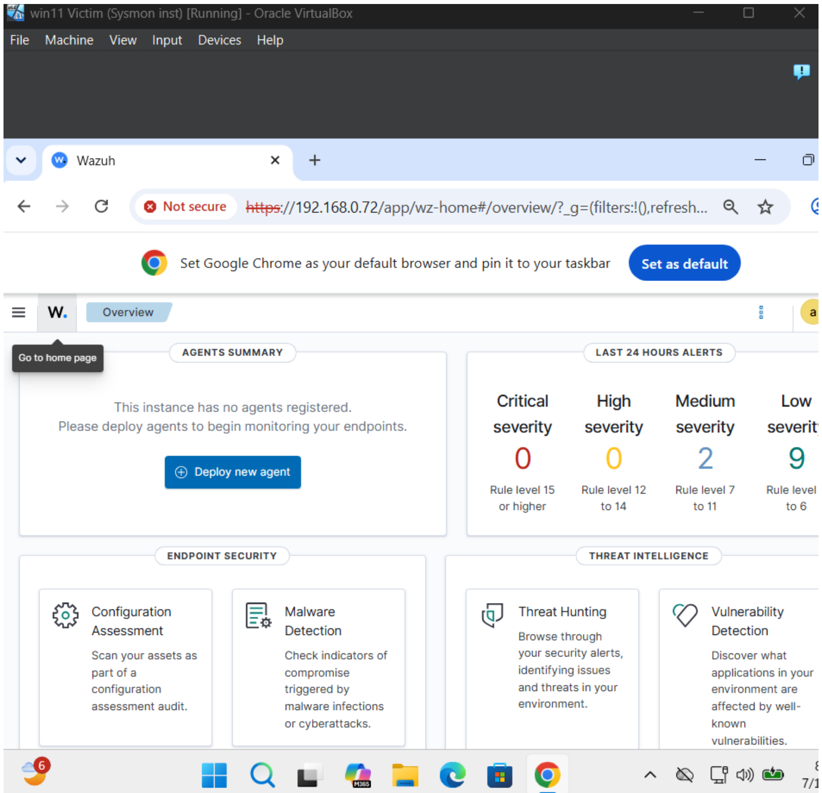


Figure 6. API connection has been achieved, dashboard currently displays 0 agents being deployed to the Windows 11 VM.

Deployment Verification

The Windows 11 endpoint was successfully enrolled into the Wazuh manager. After installing and starting the Wazuh Agent service, the endpoint appeared in the Wazuh Dashboard with an active status, confirming successful communication between the endpoint and the manager. This completed deployment phase and established the foundation for the centralized endpoint monitoring and security event collection.

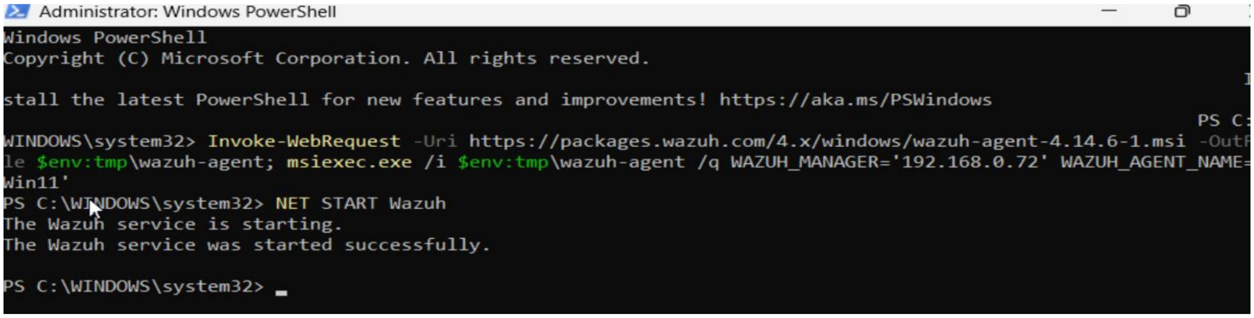


Figure 7. PowerShell executed as administrator in order to install Wazuh agent on Windows 11 endpoint.

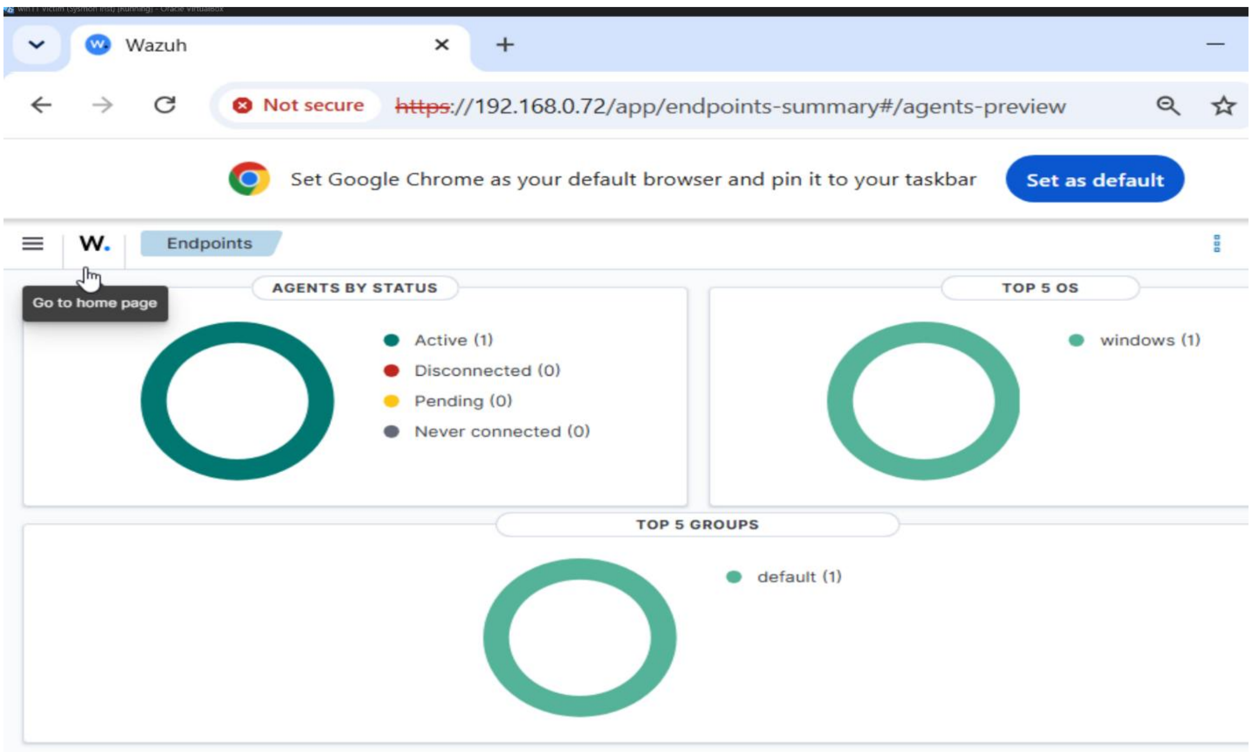


Figure 8. Wazuh manager dashboard displays successful installation of (1) active agent

and connection with (1). Windows 11 OS endpoint.

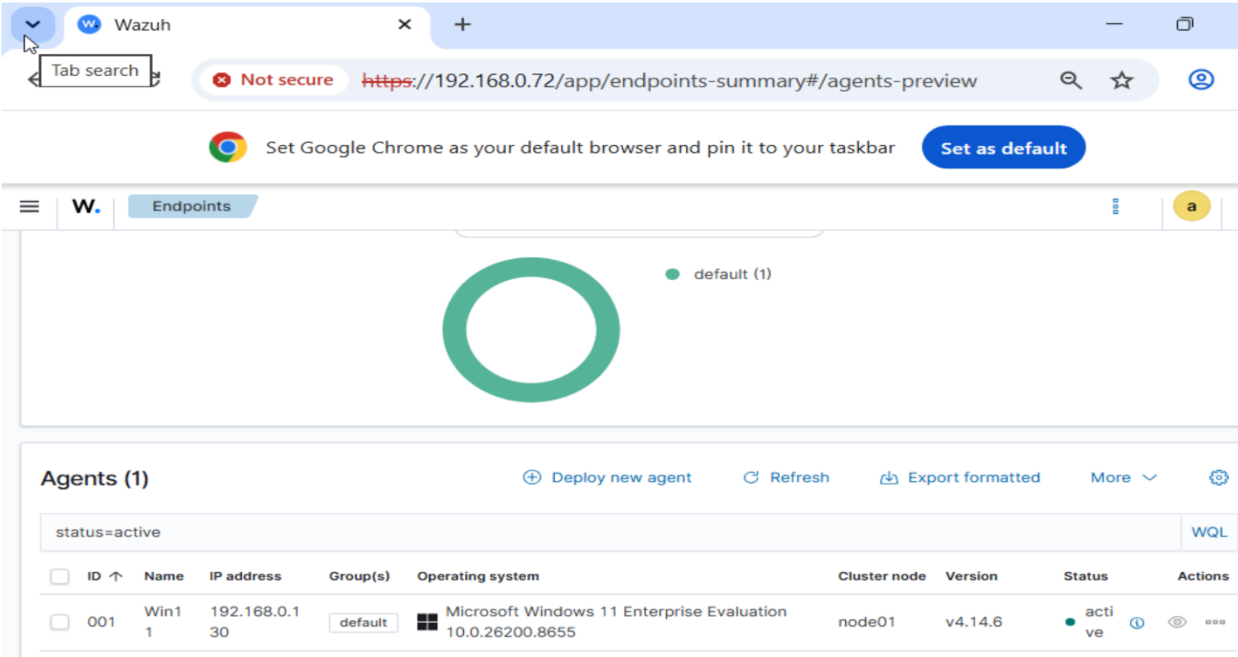


Figure 9. Installed Wazuh Agent correctly displays information on endpoint such as the IP address and OS of Windows 11 VM.